

Bank Customer Segmentation for Loan Campaign

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The bankload dataset contains 5,000 customer records and 14 variables, capturing demographic, financial, and banking information. Key attributes include age, years of work experience, income, family size education level, and mortgage value. Additional fields record average monthly credit card spending (CCAvg), indicators for personal loans, security and CD accounts, online banking usage, and credit card ownership. The dataset provides a comprehensive profile for customers, enabling analysis of financial behavior and loan adoption patterns. This information supports clustering segmentation strategies useful for targeted marketing and lending decisions. Figure 1 captures the first five rows of the dataset ensuring that the bankloan file was uploaded correctly. Figure 2 provides the informational summary of that data.

Figure 1

First Five Rows of the Bankloan Dataset

```
[1]: import numpy as np, pandas as pd
import matplotlib.pyplot as plt, seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

C:\Users\schw\anaconda3\envs\ds-marketing\lib\site-packages\sklearn\utils\fixes.py:28: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to Setuptools<81.
  from pkg_resources import parse_version # type: ignore

[2]: bank = pd.read_csv("bankloan.csv")
bank.head()
```

	ID	Age	Experience	Income	ZIP Code	Family	CCAvg	Education	Mortgage	Personal Loan	Securities Account	CD Account	Online	CreditCard
0	1	25	1	49	91107	4	1.6	1	0	0	1	0	0	0
1	2	45	19	34	90089	3	1.5	1	0	0	1	0	0	0
2	3	39	15	11	94720	1	1.0	1	0	0	0	0	0	0
3	4	35	9	100	94112	1	2.7	2	0	0	0	0	0	0
4	5	35	8	45	91330	4	1.0	2	0	0	0	0	0	1

Note. This figure displays the initial five rows of the dataset used in Activity 3.01. The data includes customer demographic and financial variables.

Figure 2

Information Summary

```
[3]: bank.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype  
---  --
0   ID                     5000 non-null   int64   
1   Age                    5000 non-null   int64   
2   Experience              5000 non-null   int64   
3   Income                  5000 non-null   int64   
4   ZIP Code                5000 non-null   int64   
5   Family                  5000 non-null   int64   
6   CCAvg                   5000 non-null   float64  
7   Education               5000 non-null   int64   
8   Mortgage                5000 non-null   int64   
9   Personal Loan           5000 non-null   int64   
10  Securities Account      5000 non-null   int64   
11  CD Account              5000 non-null   int64   
12  Online                  5000 non-null   int64   
13  CreditCard              5000 non-null   int64   
dtypes: float64(1), int64(13)
memory usage: 547.0 KB
```

Note. The dataset consists of 5,000 entries with 14 columns. All fields are completed with no missing values. Data types are primarily integers, with one float field.

Figure 3

Statistics of Standardized Income and CCAvg

```
[4]: scaler = StandardScaler()
bank[['Income_scaled', 'CCAvg_scaled']] = scaler.fit_transform(bank[['Income', 'CCAvg']])
bank[['Income_scaled', 'CCAvg_scaled']].describe()
```

	Income_scaled	CCAvg_scaled
count	5.000000e+03	5.000000e+03
mean	1.449507e-16	-6.394885e-17
std	1.000100e+00	1.000100e+00
min	-1.428969e+00	-1.108987e+00
25%	-7.554825e-01	-7.084116e-01
50%	-2.123482e-01	-2.506106e-01
75%	5.263146e-01	3.216407e-01
max	3.263712e+00	4.613525e+00

Note. This figure displays the statistics for the standardized values of Income and CCAvg using the StandardScaler. Both features have been transformed to have a mean near zero and a standard deviation of one. This allows for comparability in clusters. The table shows the count, mean, standard deviation, minimum, quadrants, and maximum value for each scaled feature.

Figure 4

K-Means Clustering

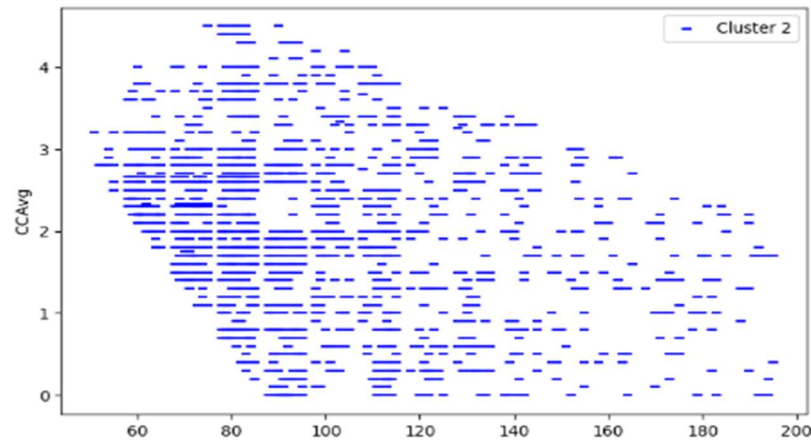
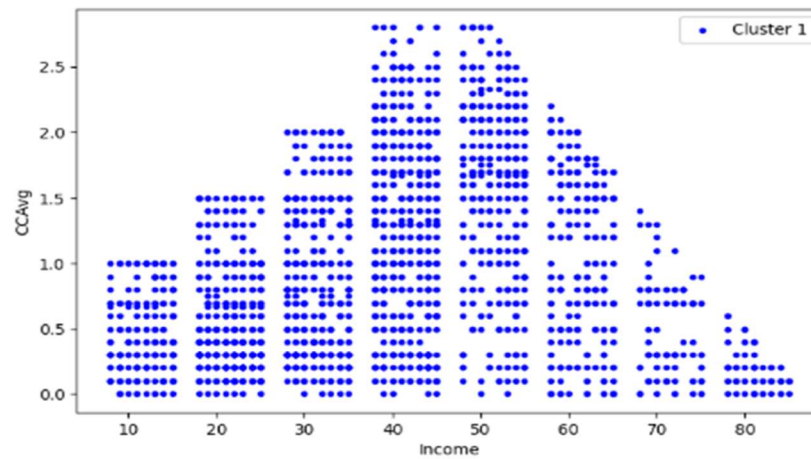
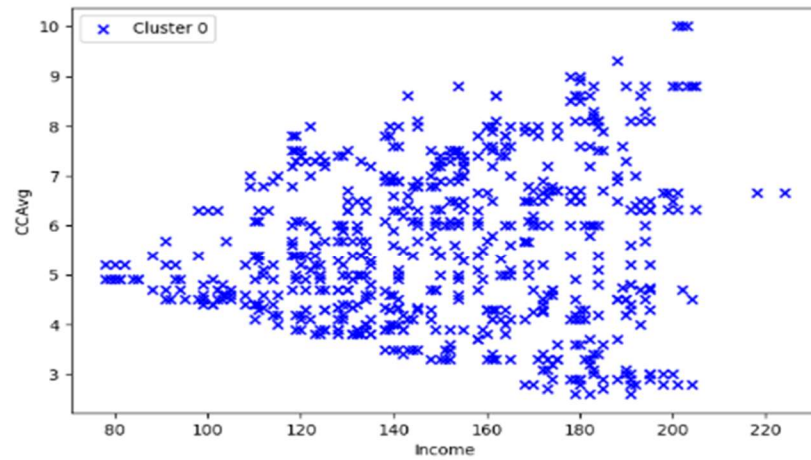
```
[5]: model = KMeans(n_clusters=3, random_state=42)
cluster_cols = ['Income_scaled', 'CCAvg_scaled']
model.fit(bank[cluster_cols])
bank['Cluster'] = model.predict(bank[cluster_cols])
```

Note. This figure shows the implementation of the k-means clustering algorithm using the scaled Income and CCAvg variables. The model was specified with three clusters and a fixed random state of 42 for reproducibility. A new column, Cluster, was added to the dataset to record each observation's assigned cluster.

Figure 5

Scatter Plots of K-Means Cluster

```
[6]: markers = ['x', '.', '_']
for clust in range(3):
    temp = bank[bank['Cluster'] == clust]
    plt.figure(figsize=(8,5))
    plt.scatter(
        temp['Income'],
        temp['CCAvg'],
        marker=markers[clust],
        color='blue',
        label=f"Cluster {clust}"
    )
plt.xlabel('Income')
plt.ylabel('CCAvg')
plt.legend()
plt.show()
```



Note. These scatter plots display the three clusters generated using K-Means clustering on standardized Income and CCAvg values. Each chart represents one cluster. They are shown in blue with different marker styles distinguishing the group.

Figure 6

Average Income and CCAvg by Cluster

```
[7]: bank.groupby('Cluster')[['Income', 'CAvg']].mean()
```

```
[7]:
```

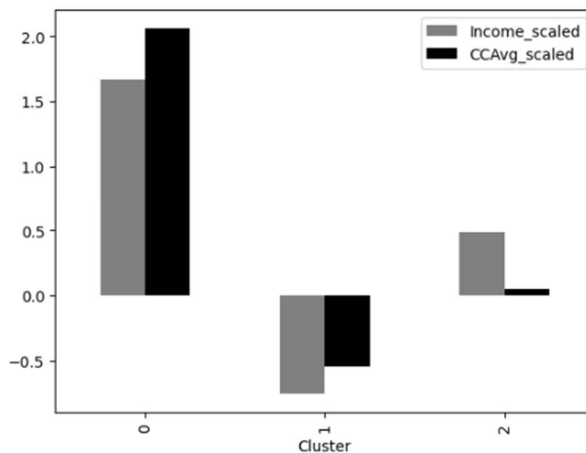
	Income	CAvg
Cluster		
0	150.390282	5.540345
1	39.135072	0.982417
2	96.142777	2.025165

Note. The table shows the mean values of Income and CCAvg for each of the three clusters. Cluster 0 represents the customers with the highest average income and credit card spending. Cluster 1 has the lowest average. The results illustrate the segmentation of customers based on financial characteristics.

Figure 7

Comparison of Income and CCAvg Across Clusters

```
[8]: bank.groupby('Cluster')[['Income_scaled', 'CAvg_scaled']].mean().plot.bar(color=['gray', 'black'])
plt.show()
```



Note. This chart compares the scaled mean values of Income and CCAvg across the three clusters. Cluster 0 shows the highest averages for both variables while Cluster 1 shows the lowest.

Figure 8

Cluster Averages for Demographic, Financial, and Loan Features

```
[9]: sel_cols = ['Income', 'CCAvg', 'Age', 'Mortgage', 'Family', 'CreditCard', 'Online', 'Personal Loan']
bank.groupby('Cluster')[sel_cols].mean()
```

```
[9]:
```

	Income	CCAvg	Age	Mortgage	Family	CreditCard	Online	Personal Loan
Cluster								
0	150.390282	5.540345	43.924765	89.537618	2.017241	0.285266	0.605016	0.413793
1	39.135072	0.982417	45.797197	42.231218	2.513429	0.294667	0.592448	0.000000
2	96.142777	2.025165	45.184049	65.185165	2.363636	0.296152	0.600112	0.120468

```
[10]: from datetime import datetime
current_dateTime = datetime.now()
print(current_dateTime)

2025-09-28 13:52:31.075183
```

Note. This table presents the average values of selected demographic and financial features across the three clusters. Cluster 0 shows the highest income, mortgage, and credit card spending. It also has the highest propensity for taking a personal loan with approximately 41% of its members holding a loan. Cluster 2 has a moderate uptake at around 12% while Cluster 1 has no loan adoption. These results suggest that higher income, higher spending customers (Cluster 0) are the most responsive segment for personal loan products.

Summary

The clustering analysis of the bankloan dataset provides meaningful insights into customer segmentation and loan adoption patterns. Using Income and CCAvg as primary clustering variables, three distinct groups emerged. Cluster 0 represents higher income and higher spending customers. They hold the largest proportion of personal loans at 41% suggesting they are the most receptive audience for personal loan products. This aligns with their greater financial capacity and demonstrated willingness to borrow. Cluster 2 occupies the middle ground with moderate income and credit card spending with a smaller but notable share of loan holders at 12%. This group may represent an untapped opportunity for the bank if targeted with offers or more flexible loan terms. In contrast, Cluster 1 consists of lower income and lower spending customers, none of whom currently hold a personal loan. This group may not be suitable for direct loan marketing since their financial profile suggests limited capacity or interest in loan adoption.

For the bank, these findings highlight the value of segmentation for optimizing marketing resources. By focusing on Cluster 0 and 2, the bank can maximize loan uptake and profitability

while avoiding wasted efforts on low probability segments (Cluster 1). Understanding the characteristics of each cluster allows for customized marketing strategies such as premium loan offers for Cluster 0, education or introductory products for Cluster 2, and non-loan products for Cluster 1. The analysis demonstrates how clustering supports data driven decision-making in financial marketing.